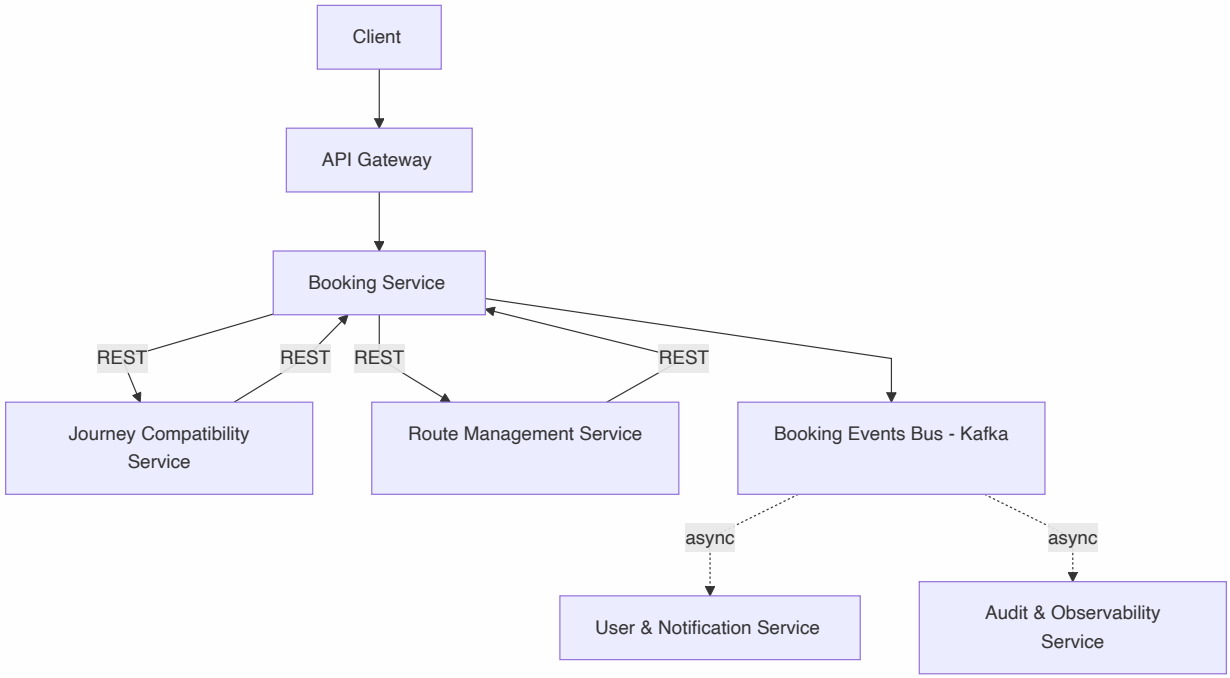
1. System Architecture Overview



2. Services

Service	Responsibility	Owner
API Gateway	Single entry point. JWT authentication, rate limiting, routing, circuit breaking.	Shreyansh Soni
Booking Service	Transactional core. Booking creation/cancellation, idempotency enforcement, outbox-based event publishing.	Naman Taneja
Journey Compatibility Service	Fast synchronous capacity validation using Redis atomic counters. Falls back to DB read replica on Redis failure.	Amar Singh
Route Management Service	Stores route definitions and capacity limits. Data changes infrequently; aggressively cached.	Eshan Das
User & Notification Service	Stores user profiles. Consumes booking events asynchronously and dispatches email/SMS/push notifications.	Gaurav Naudiyal
Audit & Observability Service	Consumes audit events from Kafka. Append-only immutable log for enforcement verification and operational monitoring.	Nkem Mogbo

3. Third-Party Technologies

Component	Technology	Purpose
Frontend	React Typescript	Frontend / Client
API Gateway	Golang / Java	API Gateway implementation
Core Services	Spring Boot (Java)	Booking, Route, Notification services
Database	PostgreSQL	Persistent relational storage (per-service)
Cache	Redis	Atomic counters, per-service caching
Event Streaming	Apache Kafka	Async decoupling between services
Orchestration	Docker / Kubernetes	Containerisation and horizontal scaling
Metrics / Dashboards	Grafana	Operational monitoring and alerting
Distributed Tracing	OpenTelemetry	End-to-end request tracing
Cloud Platform	Microsoft Azure	Hosting, managed services, multi-AZ deployment